1. Diseases and Immunity

1.1. Pathogens

- **Pathogen:** a disease-causing organism.
- **Transmissible Disease:** a disease in which the pathogen can be passed from one host to another.
- The pathogen for a transmissible disease may be transmitted either:
  - **Direct Contact:** the pathogen is passed directly from one host to another through blood, body fluids, semen, etc.
  - **Indirect Contact:** the pathogen leaves the host and is carried in some way to another e.g., contaminated surfaces/food, from animals, from air

1.2. Body Defences

The human body has many natural defences against pathogens.

- **Mechanical barriers:**
  - Nostrils contain hairs that help trap dust
  - The skin has a thick outer layer of dead cells
- **Chemical barriers:**
  - Sticky mucus which can trap pathogens
  - In the stomach, hydrochloric acid is secreted, which kills many of the bacteria in food
- **Cells:** Pathogens that manage to get through all these defences are destroyed by white blood cells:
  - Some of these cells take in and digest the pathogens by phagocytosis
  - Others produce antibodies that incapacitate or kill the pathogen and vaccination against disease helps antibodies to be made very quickly

Ways of Controlling the Spread of Diseases

- a clean water supply
- hygienic food preparation
- good personal hygiene
- waste disposal
- sewage treatment

1.3. Active and Passive Immunity

Antibody: proteins that bind to antigens, directly destroying or marking pathogens for destruction by phagocytes.

- The surface of the pathogen contains **antigens,** and they have specific shapes.
- Specific antibodies have **complementary shapes** which fit specific antigens.
- To destroy a pathogen, antibody molecules must be made that are exactly the right shape to fit into antigens outside the pathogen.
- Antibodies lock onto antigens, leading to the destruction of pathogens/marking of pathogens for phagocytes to engulf.
- If a pathogen enters the body, it meets many lymphocytes. One of these will recognise the pathogen and divide rapidly by mitosis.
- These lymphocytes then secrete antibodies, creating **active immunity**.

Active Immunity

Active Immunity: defence against a pathogen by antibody production in the body.

- Active Immunity is gained after infection by a pathogen or by vaccination.
- **Vaccines** immunise children against diseases caused by pathogens.
- Having a population vaccinated against a particular disease helps to achieve **herd immunity**.
- **Process of Vaccination:**
  - weakened pathogens or their antigens are put into the body
  - the antigens stimulate an immune response by lymphocytes, which produce antibodies
  - The body responds to the pathogen and makes its own antibodies. Some of the lymphocytes become memory cells, which provide long-term immunity

Passive Immunity

Passive Immunity: short-term defences against a pathogen by antibodies acquired from another individual.

- Memory cells are **NOT made** in passive Immunity
- Babies get passive immunity by breastfeeding.
  - Breast milk contains antibodies from the mother, which are passed on to her baby.
  - Useful because a young baby's immune system is not well developed; the mother's antibodies can protect it against any diseases.
- Some diseases are caused by the immune system mistakenly targeting and destroying the body's own cells, thinking they are harmful. These are called autoimmune diseases.

1.4. Cholera

- **Diarrhoea:** loss of water from watery faeces
- **Oral rehydration therapy and antibiotics** are used to cure this.
- The bacterium, "*Vibrio cholerae*", causes Cholera.
- The bacterium is transmitted through contaminated water.

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- The cholera bacterium **produces a toxin that causes the secretion of chloride ions** into the small intestine, **decreasing water potential** in the gut, water moves into the gut **down the water potential gradient**, causing diarrhoea, dehydration, and loss of salts and water from the blood.

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